CIFAR-10 Image Classification Using CNN

Baseline vs Data Augmentation Experiment Report

1. Abstract

This project implements a Convolutional Neural Network (CNN) using PyTorch to classify images from the CIFAR-10 dataset. Two experiments were conducted: a baseline CNN without data augmentation and an augmented CNN using random cropping and horizontal flipping. Both models were trained for 20 epochs and evaluated on the same 10,000-image test set. The augmented model achieved 75.11% test accuracy, compared with 72.04% for the baseline model.

2. Dataset

CIFAR-10 contains 60,000 color images of size 32 x 32 pixels belonging to 10 classes.

Dataset split	Number of images
Training	45,000
Validation	5,000
Testing	10,000
Total	60,000

Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

3. Model Architecture

Layer	Configuration
Input	3 x 32 x 32
Convolution 1	3 to 32 filters, 3 x 3
ReLU + MaxPooling	2 x 2 pooling
Convolution 2	32 to 64 filters, 3 x 3
ReLU + MaxPooling	2 x 2 pooling
Flatten	4096 features
Fully Connected	4096 to 128
Output	128 to 10 classes

The CNN contains approximately 545,098 trainable parameters.

4. Experimental Setup

Setting	Baseline	Augmented
Optimizer	Adam	Adam
Learning rate	0.001	0.001
Batch size	64	64
Epochs	20	20
Data augmentation	Disabled	Enabled

The augmented experiment used random cropping with padding and random horizontal flipping. Validation and test images were evaluated without augmentation.

5. Test Results

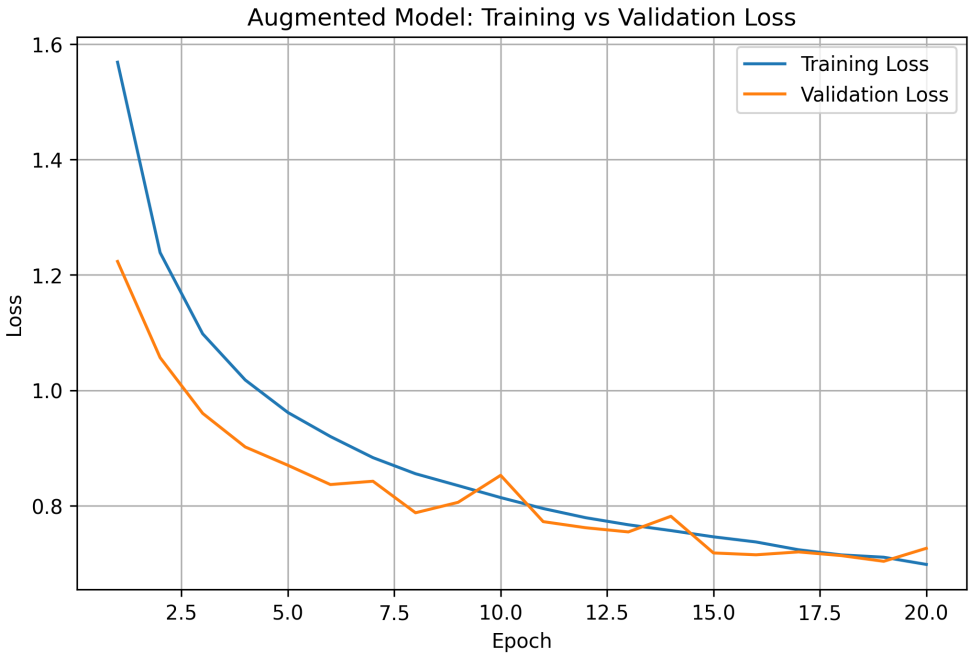
Metric	Baseline	Augmented
Accuracy	72.04%	75.11%
Macro Precision	0.7218	0.7592
Macro Recall	0.7204	0.7511
Macro F1-score	0.7201	0.7485

Test accuracy improvement: 3.07 percentage points.

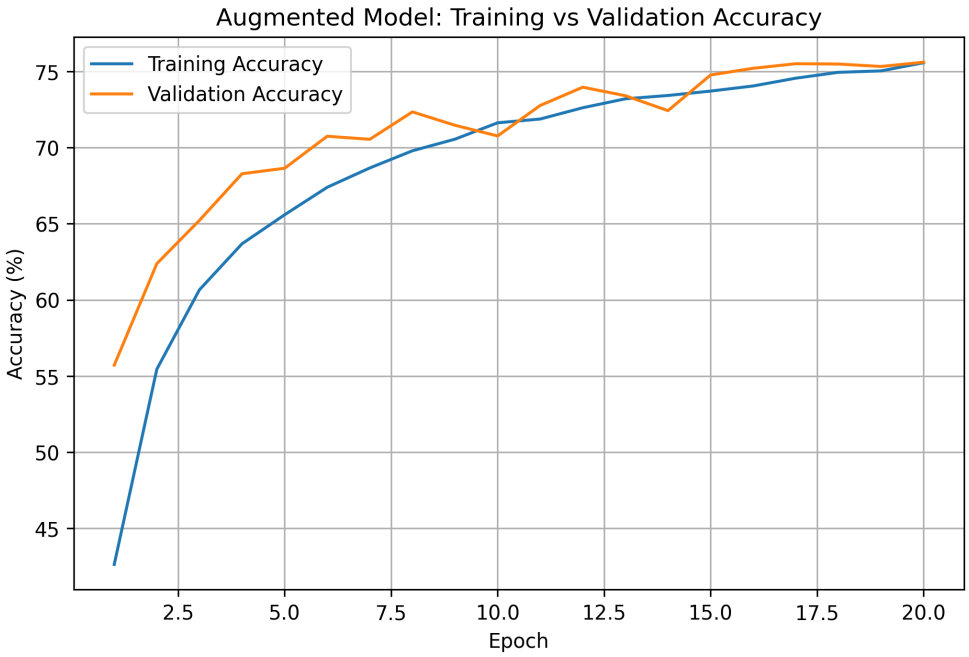
The baseline model achieved a best validation accuracy of 72.76%. The augmented model achieved 75.60%.

6. Training Visualization

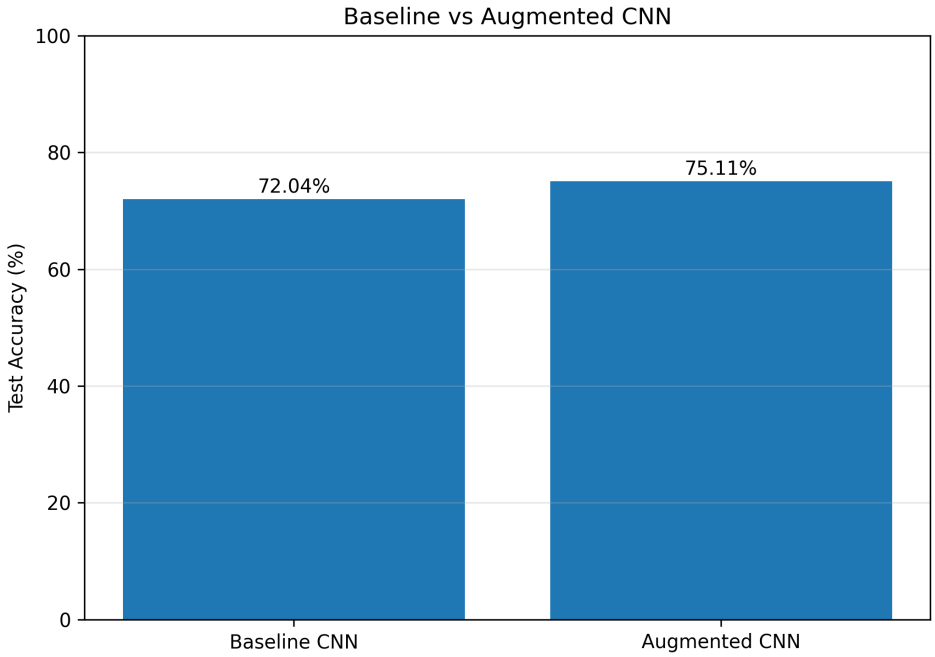
Augmented Loss Curve



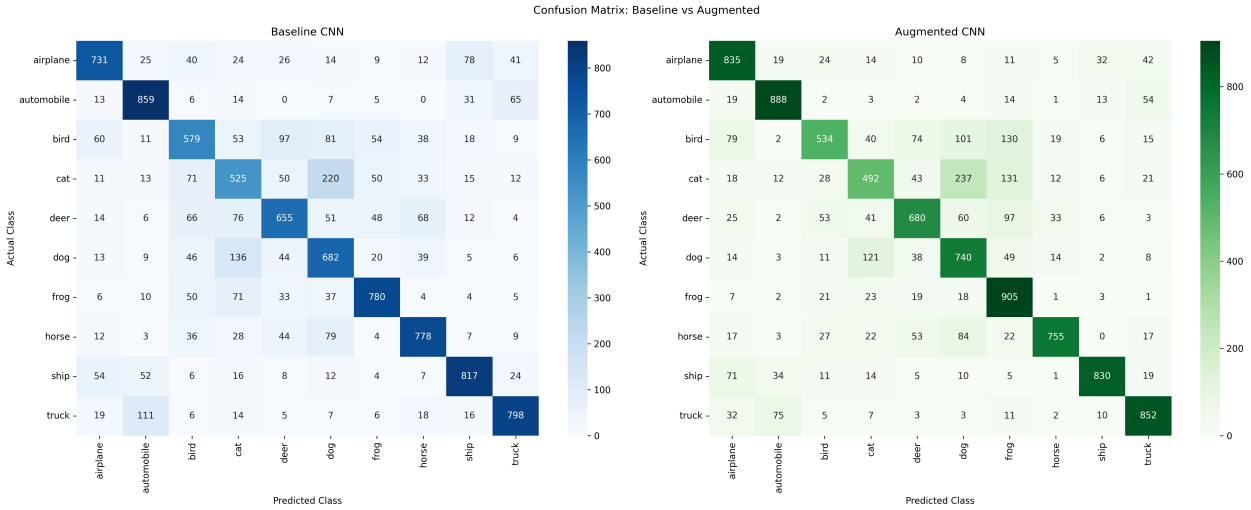
Augmented Accuracy Curve



Test Accuracy Comparison



7. Confusion Matrix Analysis



The augmented model continued to confuse cats with dogs. Cat-to-dog errors increased from 220 to 237, while dog-to-cat errors decreased from 136 to 121. Bird-to-frog confusion was also prominent in the augmented model, with 130 incorrect predictions.

8. Error Analysis

Actual class	Predicted class	Count
Cat	Dog	237
Cat	Frog	131

The average prediction confidence was 59.84% for cats incorrectly predicted as dogs and 67.40% for cats incorrectly predicted as frogs. These results show that the model can assign substantial confidence to incorrect predictions.

9. Conclusion

The augmented CNN achieved better overall test performance than the baseline CNN. Data augmentation improved test accuracy by 3.07 percentage points and improved macro precision, recall and F1-score. The baseline training results showed substantial overfitting, while augmentation helped improve generalization. However, classes such as cats and birds remain challenging. Future work could explore deeper CNN architectures, additional regularization and more advanced augmentation techniques.